



Figure 5-5: A simple program with two buttons

The user can click these buttons in any order and at any time. To handle the user's actions, the program runs in a loop and constantly checks to see if either button has been clicked. When it receives a mouse down event on a button, the program remembers that the button has been clicked and draws the depressed image of that button. When it receives a mouse up event on the button, it remembers the new state and redraws the button with its original appearance, and it plays the appropriate sound. Because the main loop runs so quickly, the user perceives that the sound plays immediately after they click the button. Each time through the loop, the program redraws both buttons with an image matching each button's current state.

Using Pygame

At first, pygame may seem like an overwhelmingly large package with many different calls available. Although it is large, there's actually not a lot that you need to understand to get a small program up and running. To introduce pygame, I'll first give you a template that you can use for all pygame programs you create. Then I'll build on that template, adding key pieces of functionality little by little.

In the following sections, I'll show you how to:

- Bring up a blank window.
- Show an image.
- Detect a mouse click.
- Detect both single and continuous key presses.
- Create a simple animation.
- Play sound effects and background sounds.